

Report on Fraud Detection Model and Analysis

Overview

This report summarizes the findings from the analysis of a dataset. The data has been pre-processed to handle missing values, and features have been engineered to facilitate meaningful insights for a Fraud Detection model.

Data Summary

Total Rows: 10,200

Total Columns: 12

Missing Data: 0.69%

Handling Missing Data

Missing data was addressed specifically in categorical columns by replacing null values with 'unknown'. This approach was taken to maintain the integrity of the analysis without making arbitrary imputations, which is crucial in areas such as fraud detection where accuracy is paramount.

Feature Engineering

To enhance the analysis, additional features were created:

Year Extraction: This allows for examining trends over multiple years.

Weekday Extraction: This enables analysis of transaction patterns on a daily basis.

Key Findings

Transaction Analysis

Day of Highest Deposits: Transactions peaked on Mondays.

Total Amount by Country: The highest recorded amount was \$2,920,032.50.

Fraud Cases

Total Fraud Cases (Pre-Resampling): There were 197 cases identified.

Country with Highest Fraud Cases: The UK had the highest number of fraud cases, followed closely by another country.

KYC Tier Analysis

The Standard KYC Tier accounted for 74.8% of the total cases.

Channel Analysis

The channel with the highest transaction amounts was Mobile, which represented 60% of total transactions.

Data Resampling

To address the imbalance in the dataset, up sampling was applied. This methodology was implemented to avoid any loss of data, ensuring that the model can learn effectively from all available instances of each class.

Model Performance

The Fraud Detection model achieved an impressive 98% accuracy on the test data, indicating strong performance in identifying fraudulent transactions.

Conclusion

The analysis highlights significant trends and characteristics within the dataset that are crucial for the development and refinement of the Fraud Detection model. The handling of missing values, careful feature engineering, and effective resampling are essential in building a robust model, as demonstrated by the high accuracy achieved. Continuous monitoring and further analysis are recommended to improve detection rates and mitigate fraud effectively.